



Cellpose-MCP + CLI: Agent-Operated Cell Segmentation

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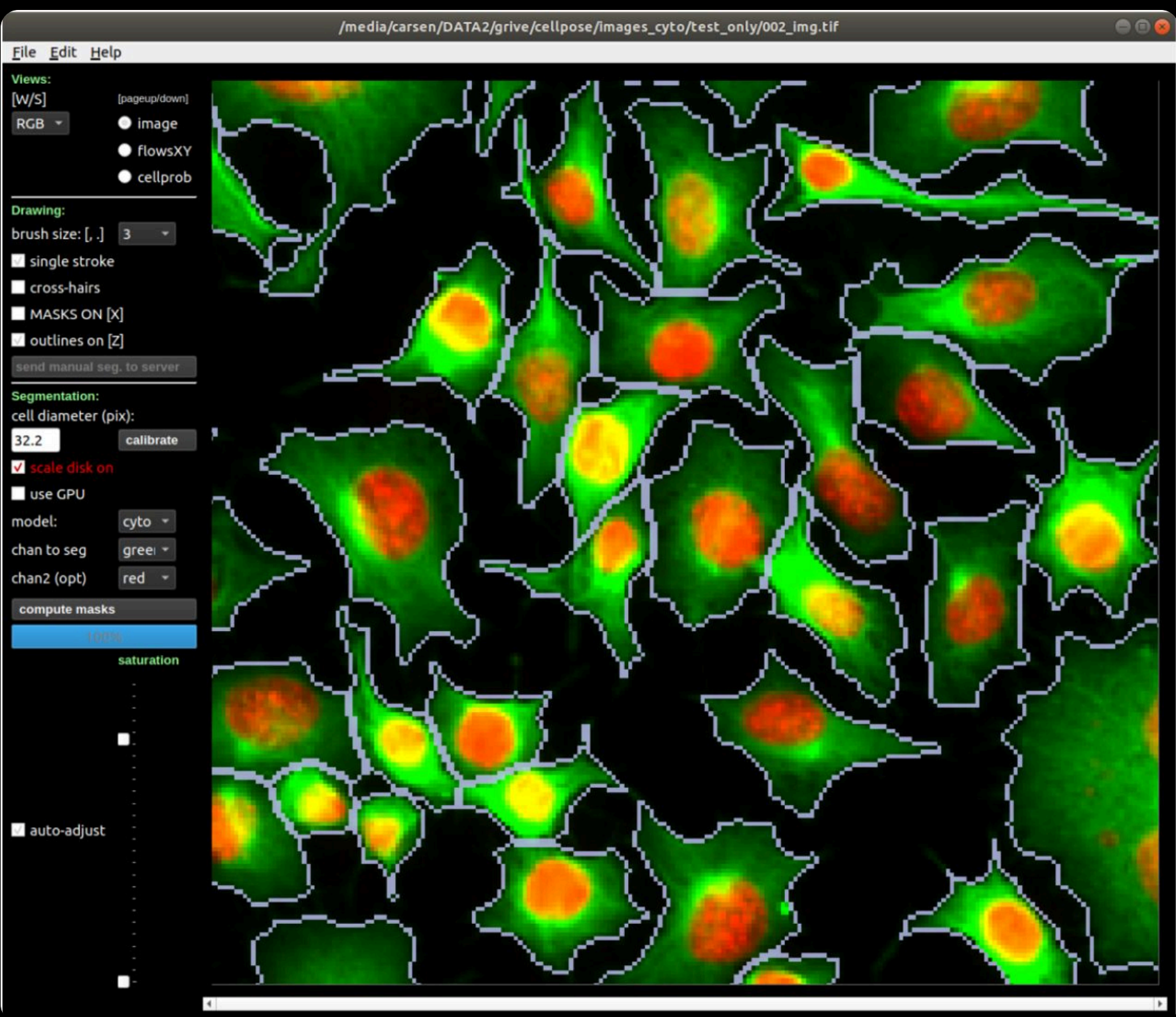
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ABSTRACT

Cellpose-mcp provides a command-line interface (CLI) and Model Context Protocol (MCP) server that let scripts and AI assistants operate Cellpose through one shared core. The package exposes 11 focused operations, including 2D/3D segmentation, batch processing, image restoration (denoising, deblurring, upsampling), and utilities. The system integrates seamlessly with Napari, enabling complete workflows from segmentation to interactive visualization. Researchers can execute sophisticated analysis pipelines by describing their goals in plain English, eliminating the need to write complex code or manually tweak Cellpose controls. By combining the generalist power of Cellpose with AI assistants, this standardizes protocols and accelerates microscopy image analysis, aiding both research and education.

CELLPOSE

Cellpose is a generalist deep learning algorithm for cellular segmentation from microscopy images. It segments cell bodies, membranes, and nuclei across diverse microscopy techniques without requiring specialized models or parameter tuning. Trained on 70,000+ segmented objects from 608 diverse images, Cellpose provides multiple pretrained models and image restoration capabilities [1].

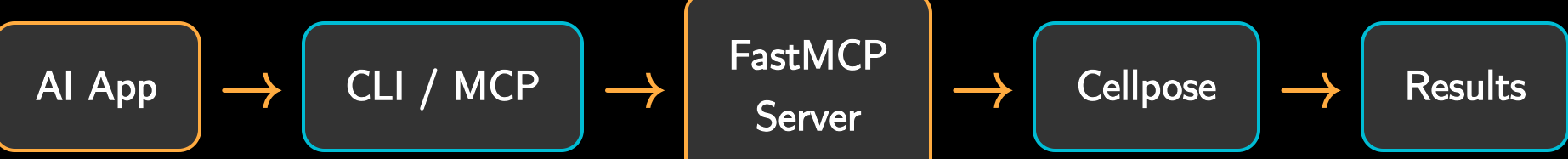


Cellpose GUI showing fluorescence microscopy segmentation with flow field overlays

CELLPOSE-MCP

Cellpose-mcp gives researchers two interfaces to the same operations: a JSON CLI for reproducible shell workflows and a local stdio MCP server for natural-language agent control. Both support segmentation, restoration, batch processing, and outputs that can be opened in visualization tools.

No Coding needed — Describe your analysis in plain English



Capabilities: cellpose-mcp exposes 11 shared operations through both CLI and MCP:

Segmentation: `segment_cells_2d` (2D images, multiple models), `segment_cells_3d` (3D volumes), `segment_cells_batch` (batch processing)

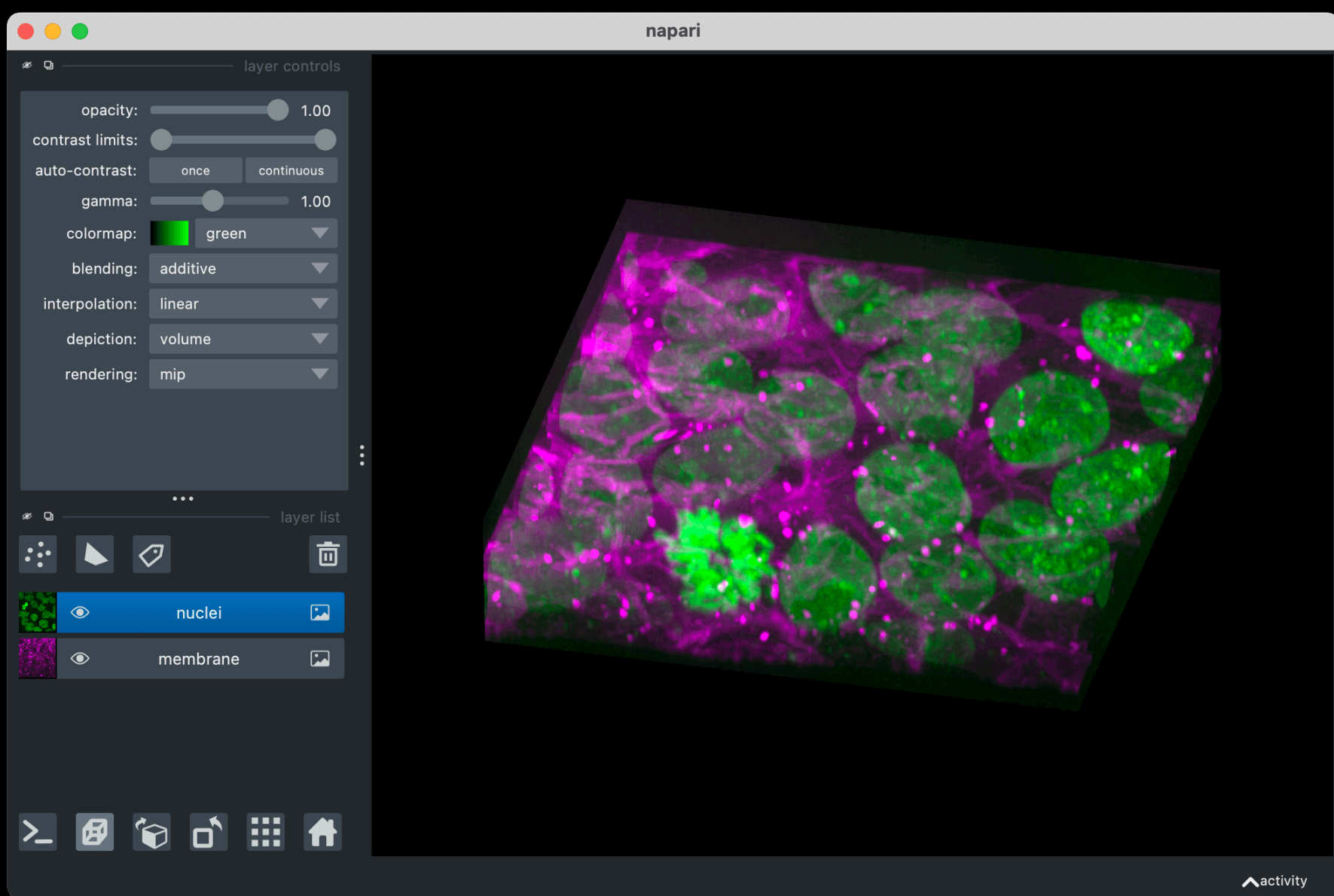
Image Restoration: `denoise_image`, `deblur_image`, `upsample_image`, `restore_and_segment` (combined pipeline)

Utilities: `list_available_models`, `estimate_cell_diameter`, `save_masks`, `load_image_info`

v0.1.5 safety: curated model identifiers, verified HTTPS model downloads, and no pickle-backed `.npy` input.

INTEGRATION WITH NAPARI

Seamless integration with `napari-mcp` [2] enables AI-driven visualization of segmentation results in the `napari` [3] viewer.



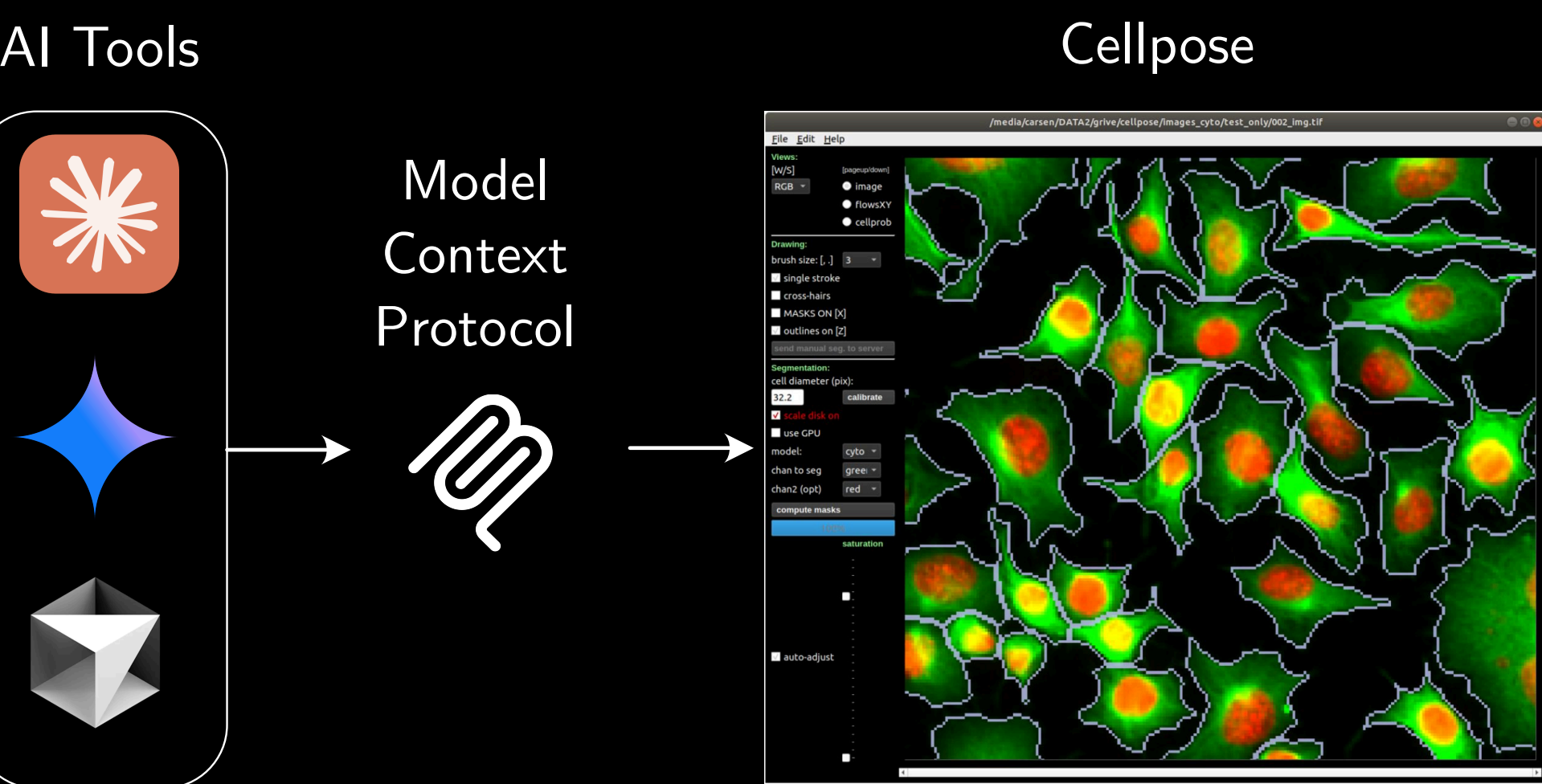
napari viewer displaying 3D multi-channel microscopy data with nuclei and membrane layers

- 1 User describes analysis in natural language
- 2 AI calls `cellpose-mcp` to segment cells
- 3 AI calls `napari-mcp` to load & visualize results
- 4 Interactive exploration in `napari` GUI

No manual file handling — immediate visual feedback — iterative refinement

CONCLUSIONS

Cellpose-mcp can transform scientific software from tools requiring technical expertise into conversational partners that amplify researcher capabilities. By combining the generalist power of Cellpose with reproducible CLI commands and modern AI assistants, we make sophisticated cell segmentation accessible to a broader scientific community. The integration with `napari-mcp` [2] enables complete workflows from segmentation to interactive visualization, reducing barriers to entry while maintaining the full power of advanced image analysis tools.



MCP enables AI-driven tool control for scientific computing

MCP (MODEL CONTEXT PROTOCOL)

The Problem

AI assistants like ChatGPT or Claude excel at text generation but cannot directly interact with software applications. For scientific researchers, this means AI cannot access standard analysis tools or proprietary data formats.

The Solution: MCP as a Universal Adapter

The Model Context Protocol (MCP) is an open-source standard developed by Anthropic that acts as a "universal remote" for AI. Instead of building custom connections for each tool, MCP provides one standardized communication language.

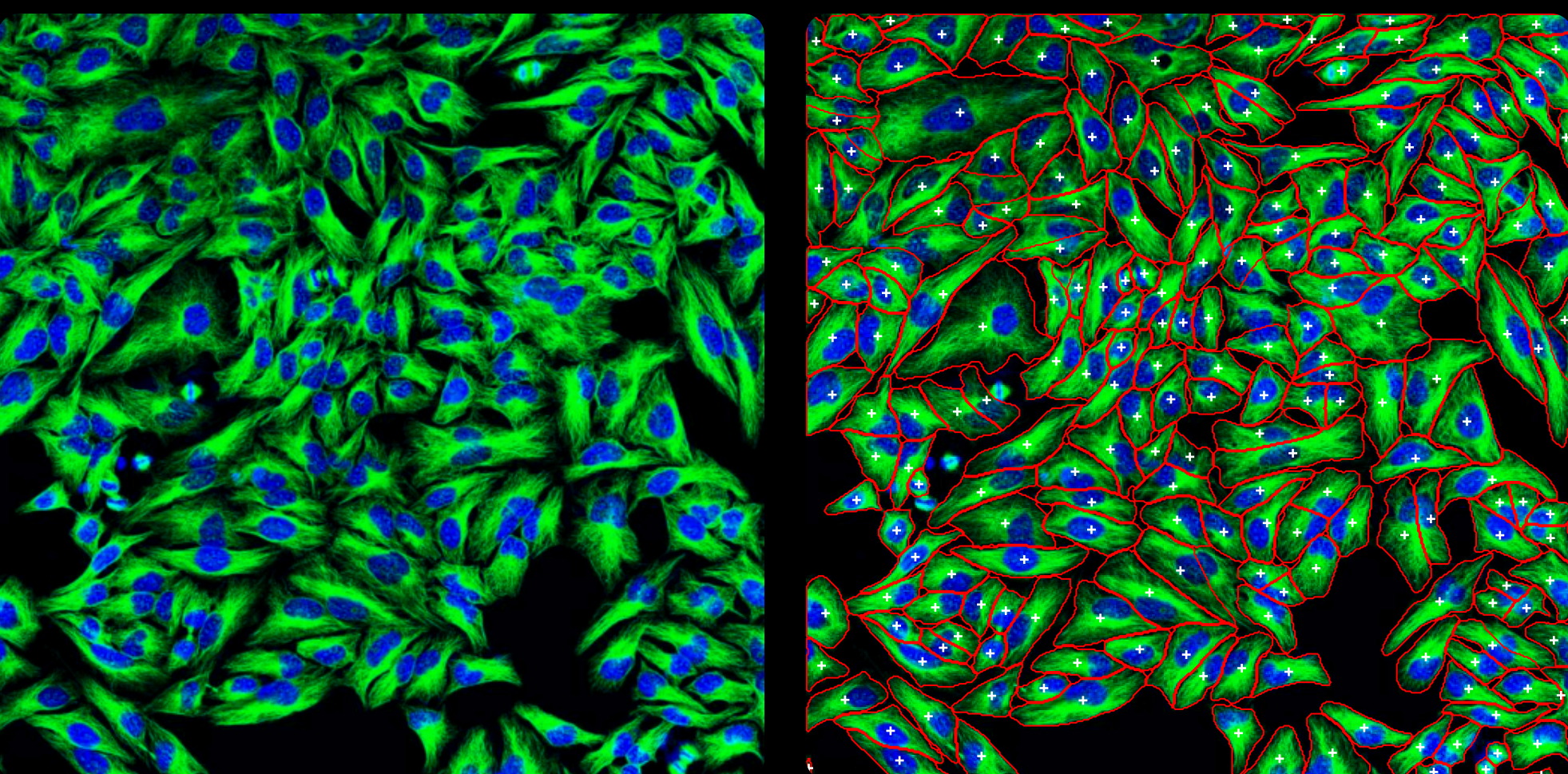
How It Works

User Request → AI Assistant (Host App) → MCP Client → Transport Layer (JSON-RPC 2.0) → MCP Server → Tool Execution → Results

EXAMPLE

User Prompt: "Segment the cells in image.tif and show the results in napari"

Input Image: image.tif - Fluorescence microscopy image with green-stained cytoplasm and blue-stained nuclei



Complete AI-driven workflow: Original fluorescence image (left) and segmented cells with annotations (right)

ADVANTAGES

- **AI Assisted:** Natural language interface enables biologists without coding experience to perform sophisticated analyses
- **Seamless Integration:** The AI agent sees what you are seeing and can help you with the analysis
- **Rapid Prototyping:** Prototype workflows conversationally without context-switching between tools
- **Local Execution:** Cellpose reads and writes local files through a stdio MCP process rather than a public service
- **Intelligent Automation:** AI suggests optimal parameters and generates batch processing scripts
- **Workflow Automation:** From raw images to quantitative analysis can be done in a one place and the AI agent has context of the entire workflow
- **Extensible:** New analysis tools can be added as MCP servers without changing user workflows, enabling incremental upgrades of scientific pipelines

FEATURES

11 Shared Operations	23 Pretrained Models
2D+3D Segmentation Images and volumes	Local JSON CLI + stdio MCP One operation core

Supported AI Applications:

- Cursor IDE
- Claude Desktop
- Codex CLI / App
- Claude Code
- VS Code

References

- [1] Stringer, C., Wang, T., Michaelos, M. & Pachitariu, M. Cellpose: a generalist algorithm for cellular segmentation. *Nat Methods* 18, 100–106 (2021). <https://doi.org/10.1038/s41592-020-01018-x>
- [2] `napari-mcp`: royerlab/napari-mcp. GitHub repository: <https://github.com/royerlab/napari-mcp>
- [3] `napari` contributors. `napari`: a multi-dimensional image viewer for python. Zenodo (2019). <https://doi.org/10.5281/zenodo.3555620>
- [5] Model Context Protocol: <https://modelcontextprotocol.io/>

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